EE3032 Introduction to VLSI—Midterm

1. (20 points) Complete the following problems.

   (a) The layer density rule should be met when you design a nano-scale IC. Why? (2 points)

   (b) What is the antenna effect? (2 points)

   (c) Draw a CMOS transistor-level schematic realizing the function $f = \overline{xy + (u+v)w}$. (4 points)

   (d) List four properties of the static CMOS logic. (4 points)

   (e) Draw a CMOS inverter and label the drain and source of the PMOS and NMOS transistors. (2 points)

   (f) List the three major components of the load capacitance of a CMOS logic gate in a CMOS circuit. (3 points)

   (g) How do you size the N and P MOS transistors of an inverter such that the output voltage of the inverter is $V_{dd}/2$ when the input voltage is $V_{dd}/2$? (3 points)

2. (20 points) If an inverter with a output load C_L is operated at f frequency under the supply voltage V_{dd}, then the switching power of the inverter can be estimated by $fC_LV_{dd}^2$. On the other hand, if an input stimulus with $t_r = t_f = \tau$ applied to an inverter of which the $\beta_n = \beta_p = \beta$, then the short-circuit power of the inverter can be estimated by $P_{sc} = k\beta(V_{dd} - V_t)^3\tau f$, where k is a constant. Also, the t_r and t_f of the inverter with a output load C_L can be estimated by $t_r = \frac{C_L}{\beta_p V_{dd}}$ and $t_f = \frac{C_L}{\beta_n V_{dd}}$.

(a) Figure 1 depicts an inverter chain driving a load C_L. Assume that $\mu_n = 2\mu_p$ and the gate capacitance of a W/L PMOS or NMOS transistor is C_g. The β_n of a W/L NMOS transistor is β. Also, the parasitic capacitance of the drain and wires is neglected. Let an unit delay be equal to $\frac{C_g}{\beta V_{dd}}$ ns. Calculate the dynamic power (switching and short-circuit power) of the inverter chain if the signal waveform shown in the leftmost figure is applied to the input of the inverter chain in the switching frequency f. (10 points)

(b) For a given technology, assume that the minimum-size inverter with a W/L NMOS and a W/L PMOS transistors and the input capacitance of the inverter is C_g. Also, the delay of the minimum-size inverter driving a C_g capacitance is t_d. For breviry, we also assume that $e = 2$ and the capacitance of the drain and wires is neglected. If your boss asks you to design a buffer chain which can drive a load $C_L = 64C_g$, how do you design the inverter chain such that the delay time of the inverter chain is minimized? Draw the inverter chain and label the size of the transistors of the inverter chain. Explain your answer. (10 points)

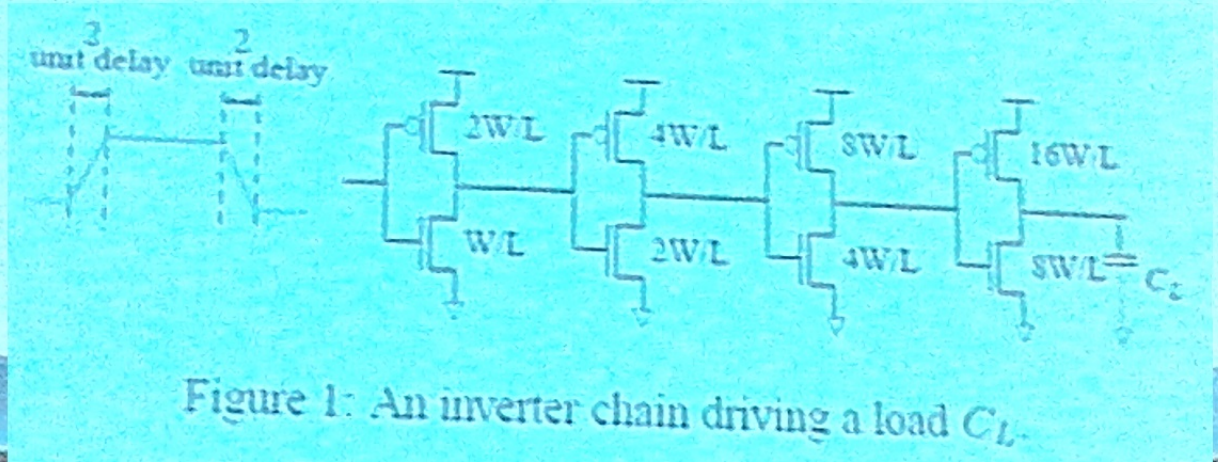


Figure 1: An inverter chain driving a load C_L.

3. (20 points) Figure 2(a) depicts a CMOS logic circuit. Assume that the delay of an inverter is 1 unit delay, where the inverter size is $W_p/L_p = 2\mu m/0.18\mu m$ and $W_n/L_n = 1\mu m/0.18\mu m$ and the input capacitance of the inverter is C_i. Also, assume that the size of each NMOS transistor and PMOS transistor of all the other gates is the same as that of the inverter. When the CMOS logic circuit is operated, possible functional inputs of the circuit are $\{I_0I_1I_2I_3I_4\}=\{01001, 01011, 11100, 10111\}$. Also, these inputs are periodically applied in sequence and the frequency is f and the power supply is V_{dd}.

(a) Assume that a dual-V_t process can be used to implement the CMOS circuit. Draw the transistor schematic diagram of the CMOS circuit in terms of the symbols shown in Fig. 2(b) such that the leakage power of the CMOS circuit is minimal without sacrificing the speed of the CMOS circuit. (10 points)

(b) Assume that the delay of each logic gate of the circuit shown in Fig. 2(a) is the same. Does the circuit exist glitches? Explain your answer. (10 points)

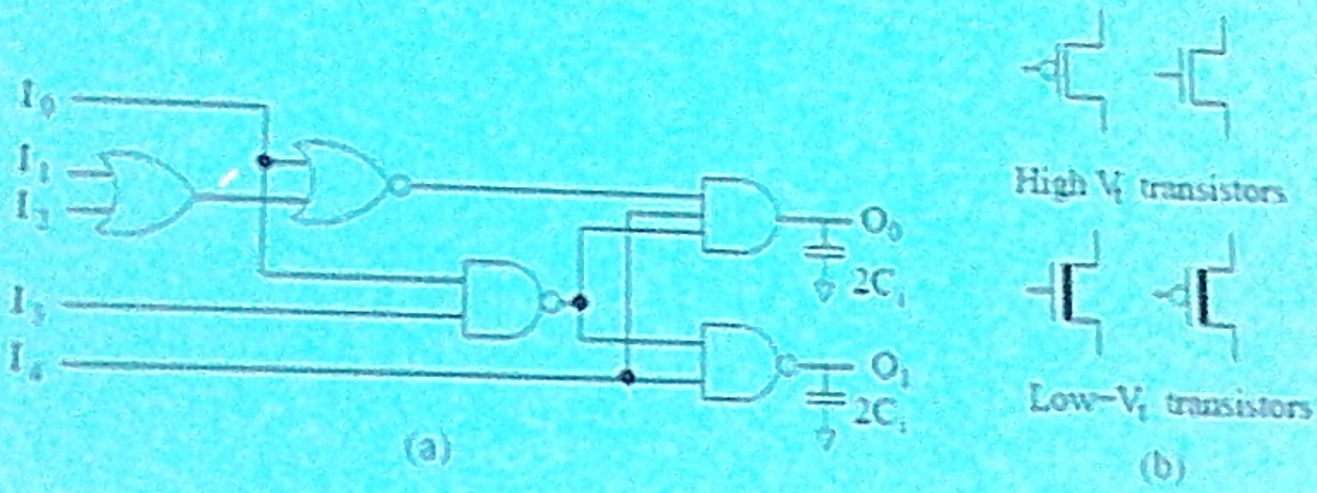


Figure 2: (a) A CMOS circuit. (b) Symbols of high-V_t and low-V_t transistors.

4. (20 points) Figure 3 shows a pass transistor logic circuit. Assume that the supply voltage $V_{dd} = 1.8V$ and the threshold voltage of the NMOS transistor is $0.4V$.

(a) Draw the corresponding waveforms at the output Y and Z if the signal waveform shown in the leftmost figure is applied to the input X. Note that you should label the smallest and largest values of voltage in the waveforms. Explain your answer. (10 points)

(b) Assume that the body effect of the transistors can be neglected. Also, the signal waveform shown in the leftmost figure is applied to the input X.

i. Identify the transistor(s) which can be operated in saturation and linear regions among the transistors $m1$, $m2$, $m3$, $m4$, $m5$, and $m6$. Explain your answer. (5 points)

ii. Identify the transistors(s) which can be operated in linear region only among the transistors $m1$, $m2$, $m3$, $m4$, $m5$, and $m6$. Explain your answer. (5 points)

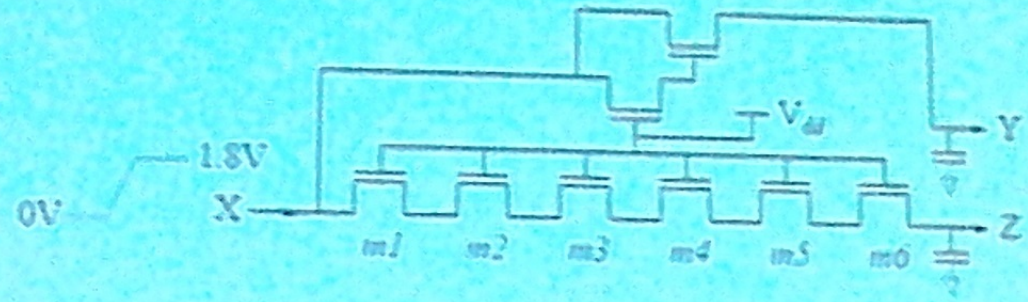


Figure 3: A pass transistor logic circuit.

5. (20 points) Figure 4 depicts interconnection RC model and different configurations of transmission gates.

(a) As Fig. 4 shows, an interconnection can be modeled as a distributed RC model. Then, we estimate the delay of an x-um interconnection by the equation kx^2, where k is a constant. What approach can be used to reduce the delay of interconnection? Explain your answer. (5 points)

(b) Assume that the delay of two cascaded transmission gates as shown in Fig. 4(b) is $8ns$. Also, the delay of a buffer which drives two cascaded transmission gates and one transmission gate as shown in Figs. 4(c) and (d) is $2ns$ and $1ns$, respectively. Complete the following questions.

i. How much is the delay of the five cascaded transmission gates as shown in Fig. 4(e)? (5 points)

ii. Redesign the five cascaded transmission gates by inserting the buffer(s) such that the delay of the circuit is smaller than $30ns$ using the smallest number of buffers. Draw the circuit and calculate its delay. (5 points)

iii. Redesign the five cascaded transmission gates by inserting the buffer(s) such that the delay of the circuit is minimized. Draw the circuit and calculate its delay. (5 points)

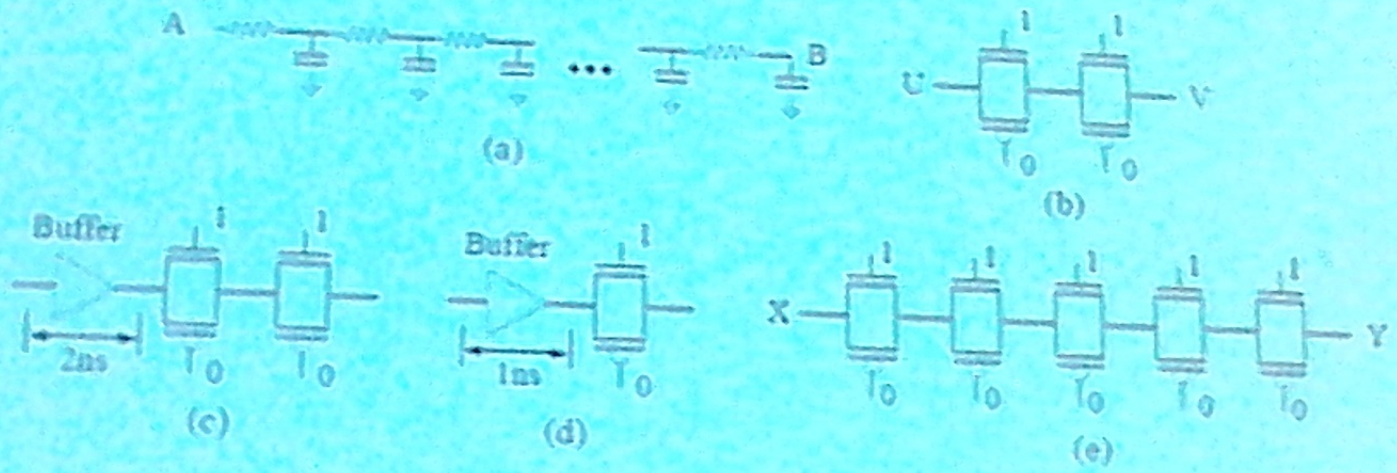


Figure 4: (a) Interconnection RC model. (b) Two cascaded transmission gates. (c) A buffer drives two cascaded transmission gates. (d) A buffer drives one transmission gate. (2) Five cascaded transmission gates.

6. (10 points) Figure 5 depicts the process cross section of a CMOS inverter.

(a) Draw the circuit level model for modeling the latchup phenomenon. (5 points)

(b) Explain the phenomenon of latchup based on the model derived in (a). (5 points)

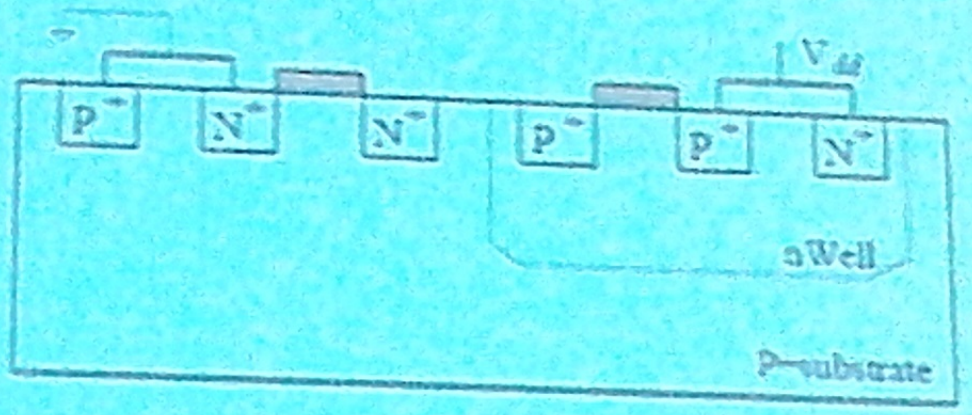


Figure 5: Process cross section view of a CMOS inverter.